

Flywheels

from zero to wheel

Mateusz "■" Winiarski

"physics", WFAIS, Jagiellonian University

today



Examples



Figure: Flywheel of extracting machine in Luiza Mine, Zabrze
(source: w.wiki/CpZ3)



Figure: Potter's wheel
(source: w.wiki/CpZ4)



Figure: Fidget spinner
(source: w.wiki/CpZ5)



Energy storage technologies

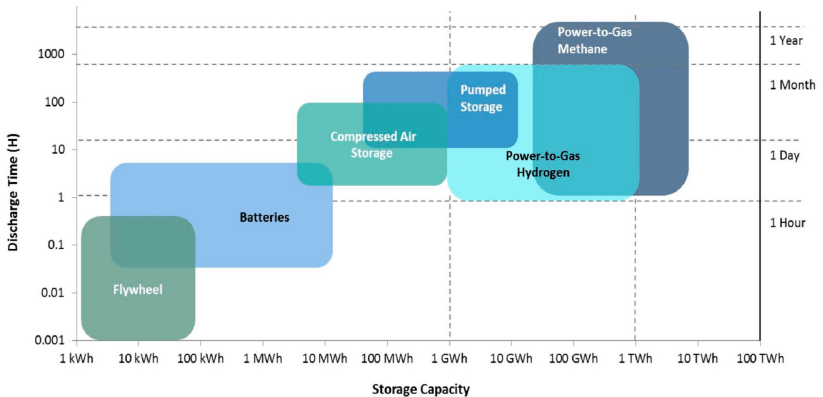


Figure: Comparison of discharge time vs capacity of energy storage technologies (source: doi.org/10.3390/en9090674)



Why flywheels are cool

Advantages:

- ▶ **High Efficiency**
- ▶ **Durability**
- ▶ **Fast Response Time**
- ▶ **Environmentally Friendly**
- ▶ **High Power Density**

Disadvantages:

- ▶ **Self-Discharge**
- ▶ **Material Constraints**
- ▶ **High Initial Cost**
- ▶ **Low Energy Density**
- ▶ **Complexity of Bearings**



Physical characteristics

moment of inertia: $J = \int_m r^2 dm$

rotational energy: $E_{kin} = \frac{1}{2} J \omega^2$



Specific energy

$$\frac{E}{m} = K \frac{\sigma}{\rho}$$

- ▶ K – shape factor [dimensionless]
- ▶ σ – tensile strength [Pa]

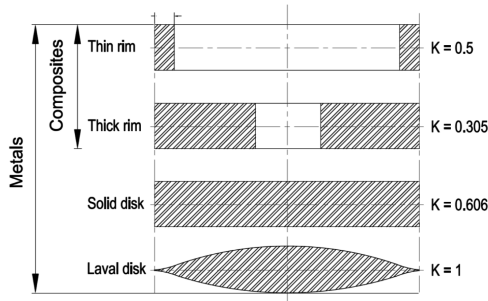


Figure: Different flywheel cross sections and their shape factors (source: doi.org/10.3390/app7030286)

Grid storage



Figure: Beacon Power's frequency regulation plant in Hazle, PA (source: doi.org/10.1109/IECON.2018.8591842)



Transportation

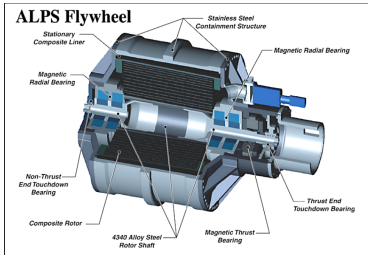


Figure: Section view of the ALPS (Advanced Locomotive Propulsion System) composite flywheel (source: doi.org/10.1109/VETECF.2003.1286244)

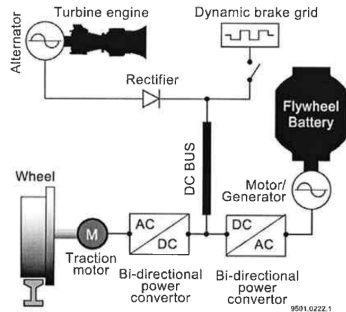


Figure: ALPS schematic diagram (source: hdl.handle.net/2152/33077)



Space applications

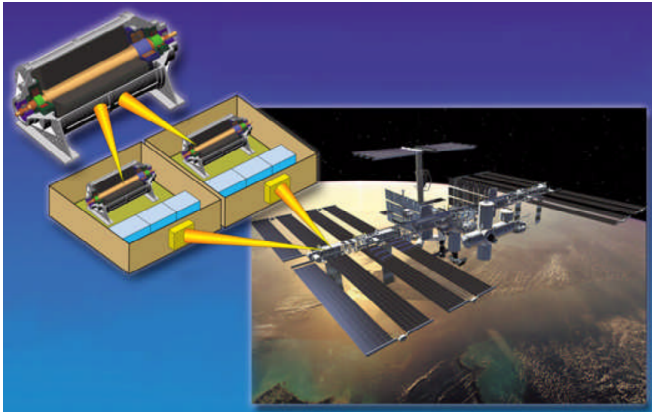


Figure: Flywheel Energy Storage System (FESS) architecture for the International Space Station (source: doi.org/10.1109/6.993788)



Comparison

Table: Flywheel applications comparison (source:
hdl.handle.net/2152/33077)

Parameter	ALPS System	Transit Bus System	NASA Space Station
Energy Stored (kW-hr)	133	2	3.6
Peak Power (kW)	2,200	150	3.6
Typical Discharge Time	3 minutes	30 seconds	30 minutes
Rotational Speed (rpm)	15,000	40,000	53,000
Machine Weight (lbs)	19,000	450	250
Motor/Generator	Induction	Permanent Magnet	Permanent Magnet
Cooling	Air and Oil	Oil and Water	Cold Plate
Bearings	Homopolar Magnetic	Homopolar Magnetic	Homopolar Magnetic
Flywheel Design	UT-CEM Cylindrical	UT-CEM Cylindrical	UT-CEM Cylindrical
Rotor Tip Speed (m/s)	950	935	920



thanks for attention



https://github.com/falcotinnunculus/przezrocza/blob/wladca/red_energy/flywheel.pdf

